

House Price Prediction

Machine Learning — Task 2

Notebook: House_Price_Prediction_Task_2.ipynb

Colab Notebook: <https://colab.research.google.com/drive/1Zndt7b4kOB28QUYiS-XVZ6sx0r2YMHZP?usp=sharing>

1. Objective

The objective of this project is to build a machine learning regression workflow to predict house prices using historical housing data. The workflow covers data preprocessing, exploratory data analysis, feature engineering, model training, model evaluation, residual analysis, prediction, and model saving.

2. Dataset Information

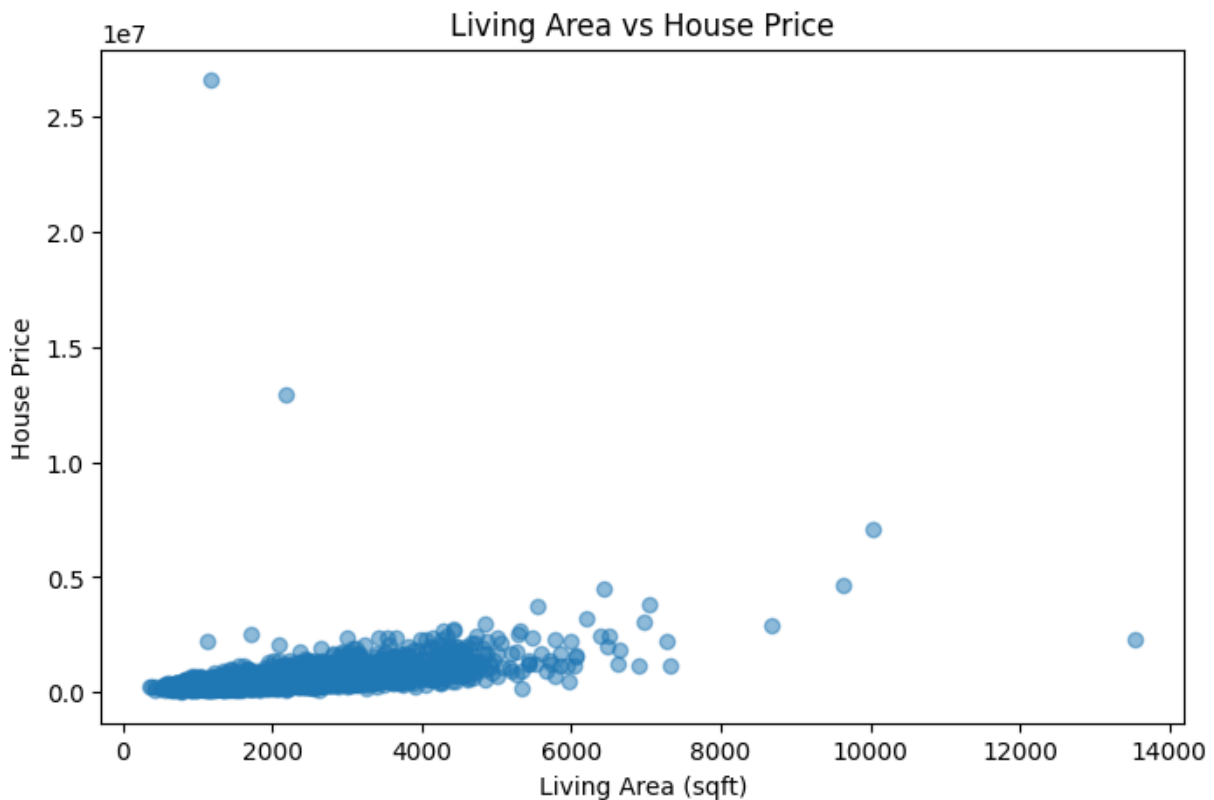
The dataset contains 4,600 records and 18 original columns. The target variable is **price**. The original data includes numerical housing attributes and categorical location/address attributes.

3. Data Preprocessing

Initial inspection found no missing values. The target variable contained 49 zero-price records. These records were removed, leaving 4,551 records and 18 columns for further processing.

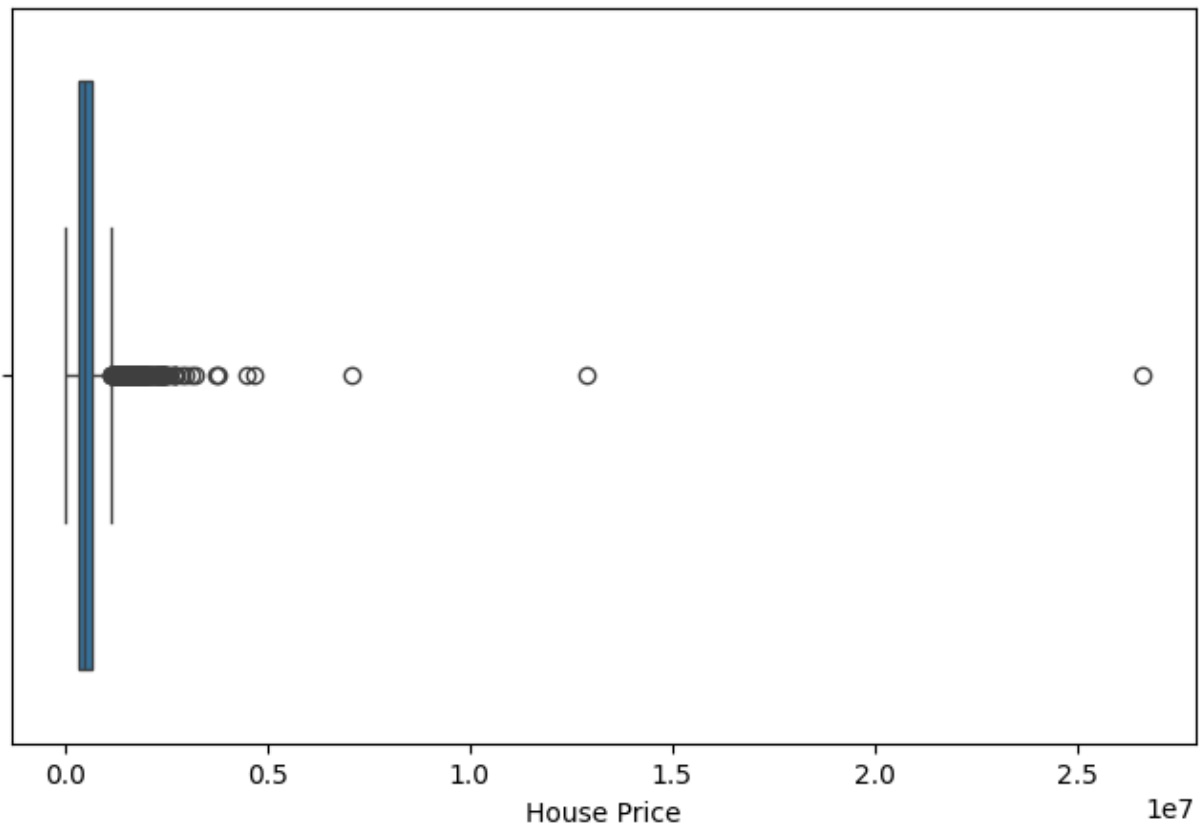
4. Exploratory Data Analysis

Four visualizations were created as required: scatter plot, box plot, histogram, and pair plot.



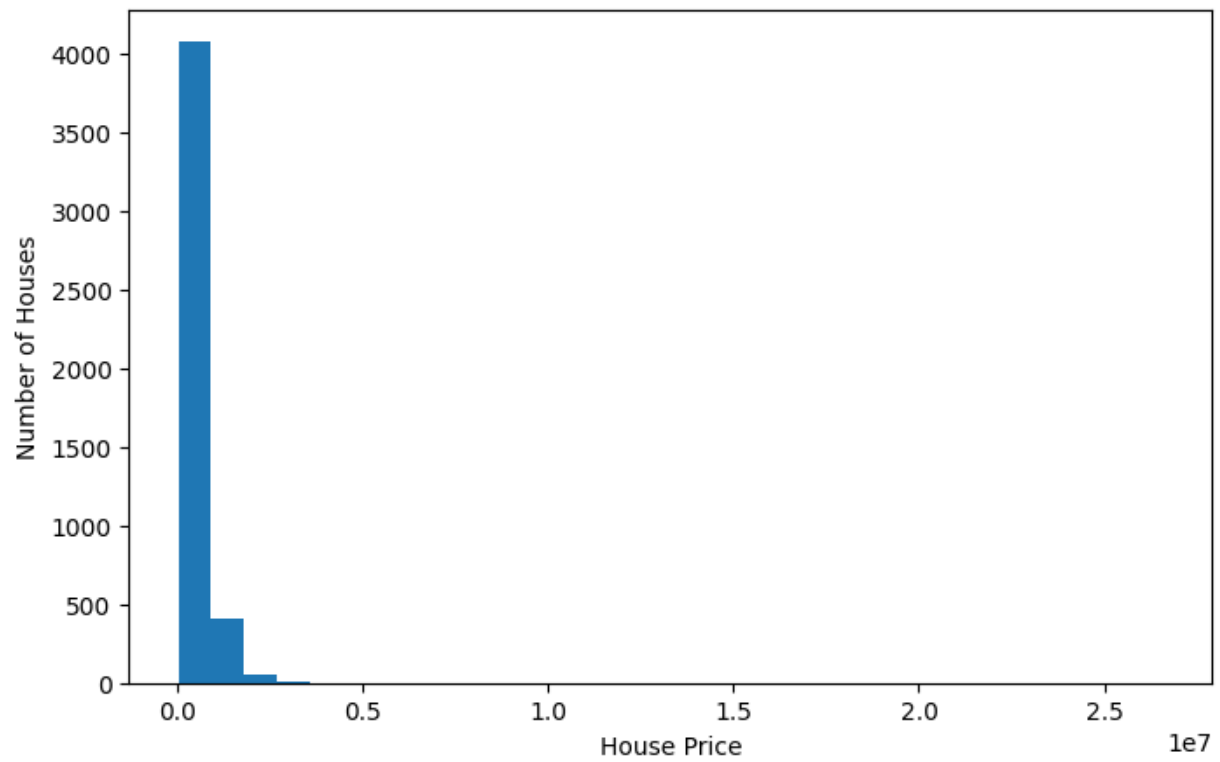
EDA — Scatter Plot

Box Plot of House Prices

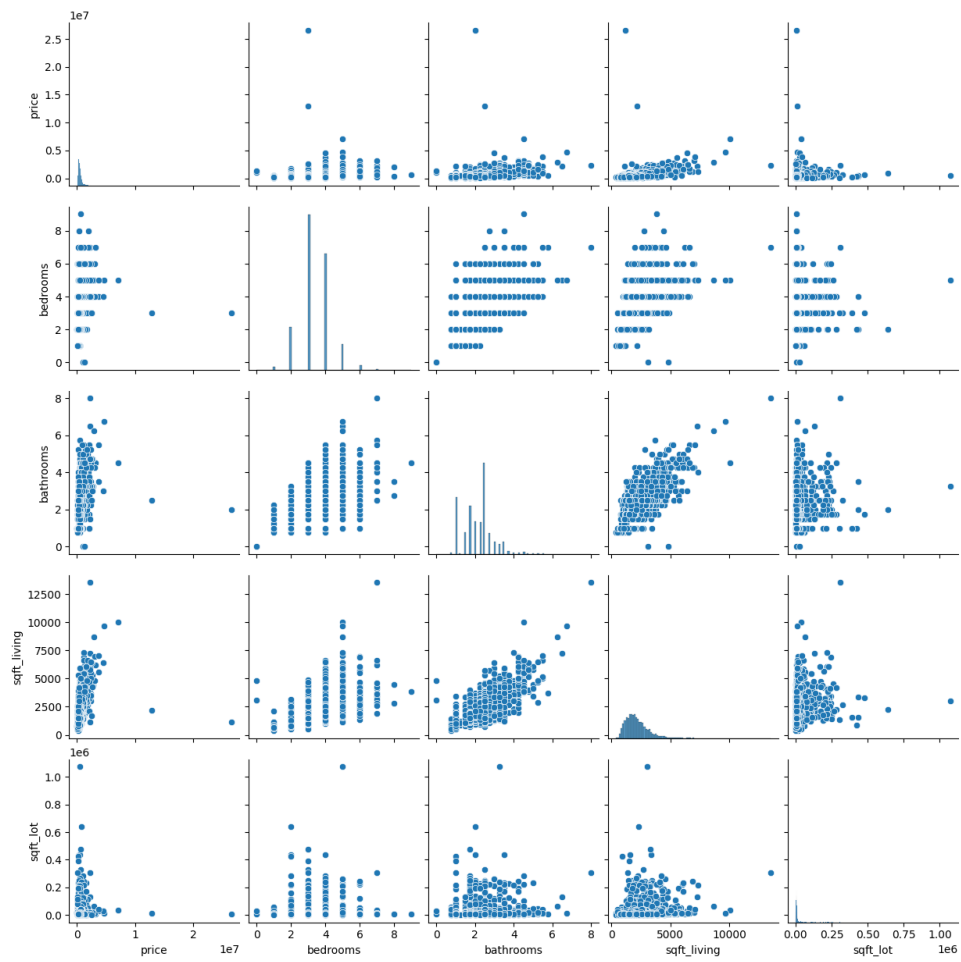


EDA — Box Plot

Distribution of House Prices



EDA — Histogram



EDA — Pair Plot

5. Feature Engineering

The date feature was transformed into **sale_year** and **sale_month**. Two additional features, **total_sqft** and **house_age**, were created. The original date column was removed. High-cardinality and constant columns were handled by removing **street** and **country**. The remaining categorical variables, **city** and **statezip**, were one-hot encoded.

Log transformation was applied to the target and selected numerical features. The transformed target used for training was **log_price**.

After one-hot encoding, the feature dataset contained 136 columns before the target/feature separation. The final feature matrix used for modelling had 137 columns.

6. Train-Test Split and Scaling

The data was split into 3,640 training samples and 911 testing samples. Feature scaling was then applied to the training and testing features.

7. Model Training and Evaluation

Model	RMSE	MAE
Linear Regression	0.257282	0.154907
Gradient Boosting	0.285705	0.195852
Random Forest	0.287697	0.187354

8. Residual Analysis

Residual analysis was performed for the Linear Regression model to inspect prediction errors.

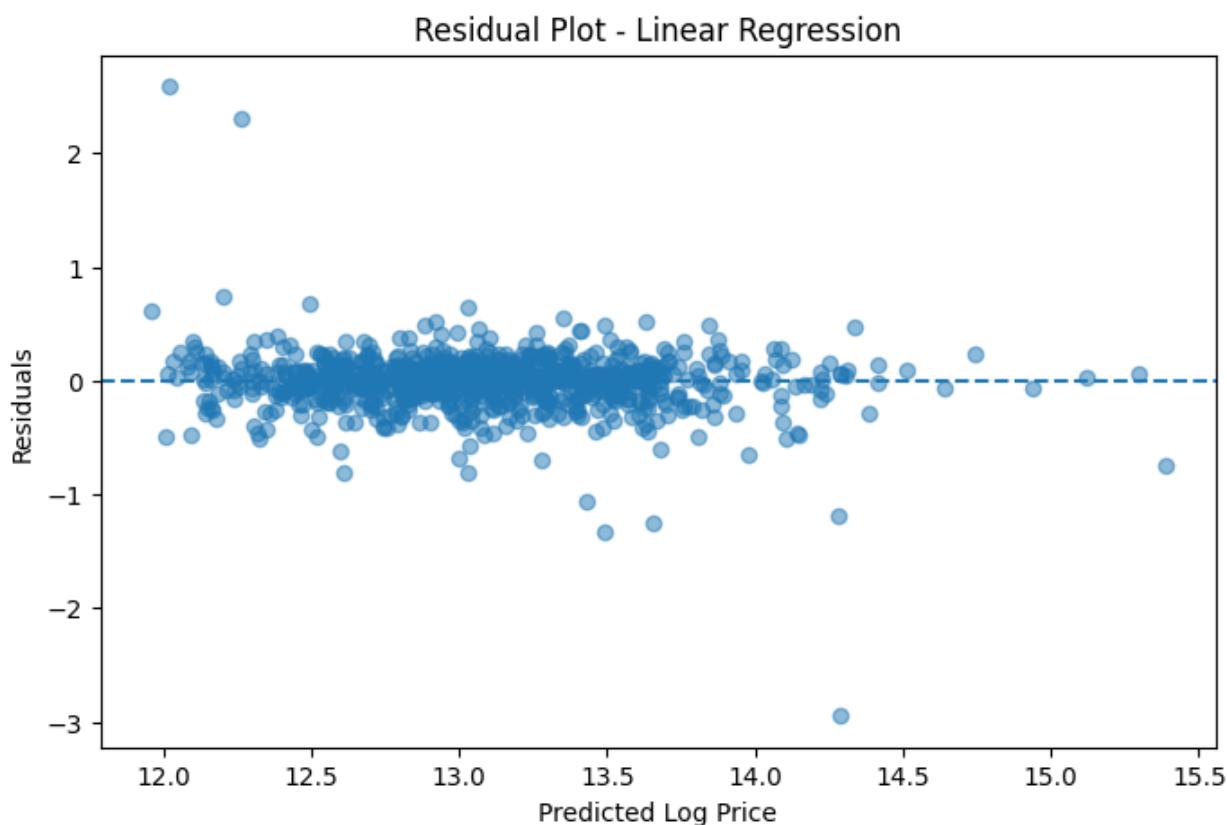


Figure: Residual analysis for Linear Regression.

9. Actual vs Predicted Price

An Actual vs Predicted visualization was created using the Linear Regression predictions converted back from log-price to the original price scale.



Figure: Actual house prices versus predicted house prices.

10. Sample Prediction

A sample prediction was demonstrated in the notebook. The prediction shown during the workflow was approximately **\$311,501.90**.

11. Saved Model and Reproducibility

The trained model and feature scaler were saved using Joblib as **house_price_model.pkl** and **house_price_scaler.pkl**. A **requirements.txt** file was also created containing pandas, numpy, matplotlib, seaborn, scikit-learn, and joblib. The saved model was successfully loaded and tested.

12. Submission Links

Colab Notebook:

<https://colab.research.google.com/drive/1Zndt7b4kOB28QUYiS-XVZ6sx0r2YMHZP?usp=sharing>

GitHub Repository: Not provided yet — add the Task 2 repository link after uploading the notebook and project files.

13. Conclusion

This project demonstrates a complete machine learning regression workflow including data preprocessing, exploratory analysis, feature engineering, categorical encoding, log transformation, model training, evaluation using RMSE and MAE, residual analysis, prediction, and model saving.